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Mobile-First Usability Guideline for Responsive E-Commerce Websites

Bimal Aklesh Kumar, Fiji National University, Fiji*

 <https://orcid.org/0000-0003-3622-7541>

ABSTRACT

Handheld devices such as smartphones, tablets, and PDAs are commonly used for online shopping. In order to satisfy web page requests on these devices, web developers are using a responsive web design approach, where websites dynamically adapt to different screen sizes and orientations. One of the challenges faced by web developers is to provide a usable experience across different devices. Motivated by this, a mobile-first usability guideline was established by extracting good usability practices from five popular responsive e-commerce websites. Validation of the usability guideline was conducted, and results indicate that it will have a positive impact on the overall user experience.

KEYWORDS

Electronic Commerce, Heuristics, Responsive Web, Usability Guideline

INTRODUCTION

Online shopping (e-commerce) has been around for a while, and it is used by many for their everyday shopping (Lissitsa & Kol, 2016; Zhou & Wang, 2014). Nowadays, it is possible to buy everything online. Thousands of e-commerce websites are available that offer a variety of products. With the advent of smartphones, tablets, and PDA's, many online shoppers are using these devices for accessing e-commerce websites (Lim, Osman, Salahuddin, Romle, & Abdullah, 2016; Mosteller, Donthu, & Eroglu, 2014). Different devices have different characteristics, such as screen size, resolution, input capability, and processing power. In order to satisfy web page requests on different devices, many developers started using responsive web design approach that allows the website to dynamically adjust to different screen dimensions. Responsive web design is now commonly used to design and develop e-commerce websites (Majid, Kamaruddin, & Mansor, 2015; Varon & Karlins, 2020). Since different devices have different characteristics, it is a challenging task to design e-commerce websites that provide usable experiences across different devices.

Motivated by this, a mobile-first usability guideline was established to support the design of responsive e-commerce websites. Mobile-first is a design philosophy where the design process starts from the smallest screen, and then the features are expanded to create a tablet and desktop version (Mullins, 2015; Van Goethem, Le Pochat, & Joosen, 2019). Usability has been identified as one of the key factors that determine the success of e-commerce websites. Hassan and Morris (2017) analyzed different attributes of twelve e-commerce websites and concluded that online customers would not return to a website if it has usability issues. Therefore in order to increase customers buying intention, providing a usable experience is mandatory. Bevan (2001) defined usability "as the ease at which

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*Corresponding Author

specified users can use the product to achieve the specified goal with effectiveness, efficiency, and satisfaction.” In the literature, there are usability guidelines available to support the design of responsive web applications. However, they are very generic and may not be very effective for designing domain-specific applications such as e-commerce. A domain-specific guideline will address issues affecting the domain (Kumar & Mohite, 2016; Kumar & Mohite, 2017; Kumar, Goundar, & Chand, 2019).

The mobile-first usability guideline was established by extracting good usability practices from five popular responsive e-commerce websites. A questionnaire was designed to guide the data extraction process. Ten usability experts used the questionnaire to assess selected e-commerce websites. Card sorting, a knowledge elicitation technique, was used to group the responses. Twenty-six usability guidelines were developed. Heuristic evaluation was conducted to validate the guideline, and the results indicate it will have a positive impact on the overall user experience. This paper is organized as follows; the background section provides information on the responsive web and prior work done in the area. The methodology section describes how the research was designed and executed. Different parts of the methodology, information extraction, guideline development, and guideline validation are described in detail. Finally, the paper concludes by discussing the work done and setting the direction for further investigation in the field.

BACKGROUND

The background section describes responsive web and prior work done in this area.

Responsive Web

World Wide Web has revolutionized the society, having a huge impact on how various sectors of the economy operate today, such as education, medical, business, and many more (Aghaei, Nematbakhsh, & Farsani, 2012; Jacksi & Abass, 2019). It has been at the forefront of the digital world. The world wide web is composed of websites that consist of multiple web pages. A web page is a hypertext document that is displayed using a web browser. Over the years, we have seen some notable technological improvements in website design and development, such as HTML, CSS, JavaScript, etc. (Duckett, 2011). Initially, the websites were viewed using desktop computers that had large screens but with the advent of handheld devices such as smartphones, tablets, and PDA's enabled with the internet, many are using these devices to assess websites. These devices have different characteristics such as screen size, input capability, etc. In order to serve websites request on devices with different screen devices, a responsive web design technique was introduced (Hussain & Mkpojiogu, 2015).

The term responsive web design was coined by Ethan Marcotte, a freelance web designer, in May 2010. It is a development paradigm that optimizes a website that easily adapts to different screen dimensions (Krzewińska et al., 2018; Nogueira, Ferreira, Carvalho, & Berreta, 2017). It provides a smooth viewing experience, navigation, and content reading. The goal of responsive web design was to use a single code base to deliver websites on multiple devices (Cosgrove, 2018; Peterson, 2014). There are many advantages of responsive web design, i) makes development faster by reducing the need to create multiple web pages for devices with different screen dimensions, ii) provides the option to degrade content if required, e.g., images may be displayed on desktop but removed from the display of smartphones, iii) provides easier navigation through panning, scrolling across different devices. There are also a few challenges of responsive web design; i) responsive web design is dependent on CSS3, and this is not supported by all browsers, ii) responsive web design works on resizing the image at times, the full image is downloaded and resized, which takes time, and space iii) making responsive websites usable across different screen sizes and resolutions.

Prior Work

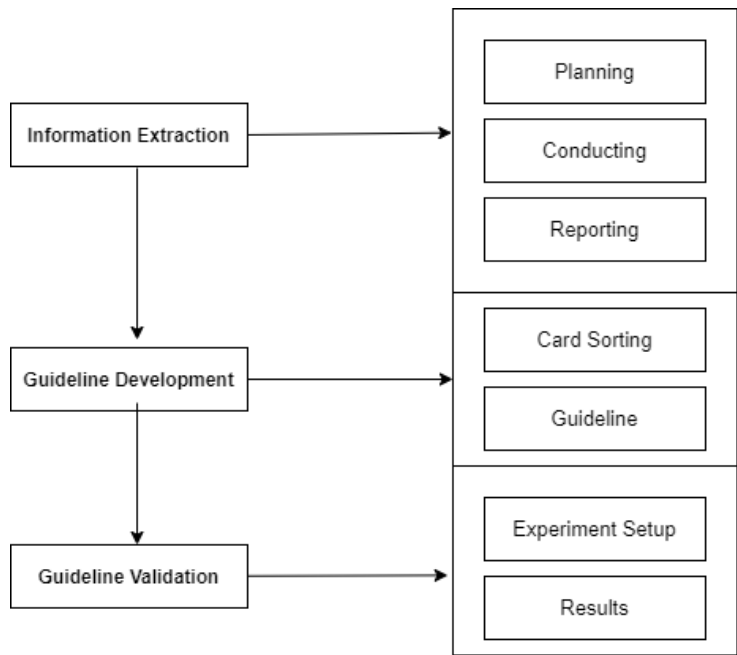
In literature, there are few guidelines to support the design of responsive web applications. Aryal (2014) studied the responsive web design paradigm and development approaches and further analyzed

these paradigms to prepare a guideline to achieve responsive design for various screen resolutions of devices. The webpage was optimized by reducing web page and image adaptability. Carver (2014) proposed user experience design patterns to support the development of responsive web applications. The design patterns are implemented using CSS. These guidelines are very generic and may not be very specific for domain-specific applications. In this study, a usability guideline was formulated specifically for e-commerce web applications.

METHODOLOGY

The methodology section was divided into three parts; Information extraction, guideline development, and guideline validation. The information extraction process includes extracting good usability practices from five commonly used websites. The guideline development includes converting the data gathered through extraction into guidelines. The guideline validation was conducted to ensure that the proposed guideline can provide a usable experience. Figure 1 illustrates the methodological process.

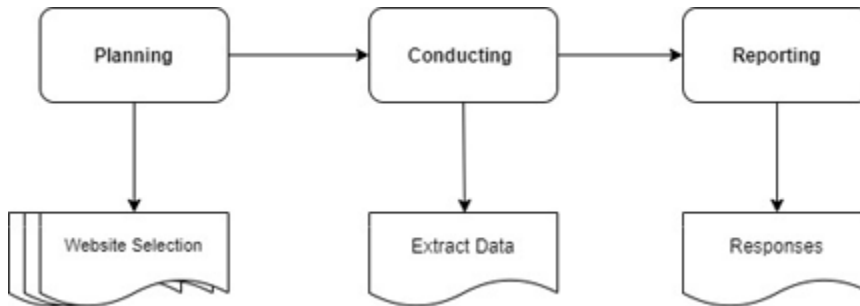
Figure 1. Methodological process



Information Extraction

The good usability practices from five popular responsive e-commerce websites were extracted. Nielsen and Molich (1990) proposed a set of usability principles referred to as heuristics which is being widely used to identify usability problems in a user interface design. If the user interface conforms to these usability heuristics, then it will provide a usable experience (Maguire, 2019). In this paper, the proposed heuristics were translated into questions and used to extract data. The process was divided into three stages; planning, conducting, and reporting. Figure 2 illustrates the information extraction process.

Figure 2. Information extraction process



Planning – this stage includes; website selection, questionnaire preparation, identifying the scope of the inspection, hardware selection, and evaluator selection.

- **Website Selection** - five popular e-commerce websites were randomly selected. These websites follow a responsive design approach.
 www.amazon.com – is an American e-commerce company and is the largest internet-based retailer in the world.
 www.ebay.com - an American multinational e-commerce company that facilitates consumer-to-consumer and business-to-consumer sales through its website.
 www.tesco.com - is a British multinational grocery and general merchandise retailer. It is the third-largest retailer in the world. It has shops in seven countries across Asia and Europe and is the market leader of groceries in the UK.
 www.walmart.com – is an American multinational retail company that operates a chain of hypermarkets.
 www.staples.com – is an American office retail company that sells office supplies and other related products.
- **Questionnaire** – was prepared to guide the data extraction process. A list of heuristics initially proposed by Nielsen and Molich (1990) were translated into questions; visibility of system status, match between system and the real world, user control and freedom, consistency and standards, error prevention, recognition rather than recall, flexibility and efficiency of use, aesthetic and minimalist design, help users recognize, diagnose, and recover from errors. Table 1 provides the lists of the questions.

Table 1. Questionnaire

ID	Heuristic
Q1	How is the visibility of the website status maintained?
Q2	How the features of the website match with the real world?
Q3	How are user control and freedom maintained?
Q4	How does the website follow consistency and standards?
Q5	What mechanisms are used to prevent errors?
Q6	What techniques are used to minimize user memory load?
Q7	How is flexibility maintained in websites?
Q8	How the website achieves aesthetic and minimalist design?
Q9	How are users able to recover from errors?
Q10	How is user support provided?

- Identifying the scope of inspection – the focus of inspection includes the commonly used features of an e-commerce website; searching for a product, viewing the product description, choosing another product, adding the product to the cart, and checkout. A task list was generated using the scope of the inspection. The task list is given below;

Task List

1. Search for Product
 - 1.1 Type in the product name of your choice in the search box, or select a category from the drop-down list. Categories might include electronics, tools, car accessories, etc.
 - 1.2 Click on the Search button, and the page displays a list of items meeting. If no product is found, refine your keyword.
2. Refine the search
 - 2.1 Once the list of products is displayed, the customer can refine the search by setting the following options
Price range
Location of item
Delivery options, etc.
3. Choose another product
 - 3.1 Choose another product, and click on the back button of the browser. To choose another product, follow step 1.1
4. View product description
 - 4.1 Click on the link of the product that you want to see the description of, and browse through the details such as the specification of the product, price, image, location of the product, supplier name, reviews, etc.
 - 4.2 Try to purchase the product, click on the buy item or add to cart link, or if you wish to place a bid, click on place a bid or make an offer link.
5. Add to Cart
 - 5.1 Enter the order quantity
 - Hardware Selection– necessary hardware devices were selected for the evaluation. Ali et al. (2014) recommend the use of actual devices instead of the simulator to obtain better results. Therefore, smartphones, tablet pc, and iPad were selected.
 - Evaluator Selection – ten usability experts were selected who have experience in heuristic evaluation. Each website was assigned to two participants.

Conducting – information extraction was carried out by the evaluator. The basic steps include; reviewing the procedure to clear any doubts, performing the tasks, and answering the questions. Prior to the beginning of the extraction process, the aim and objectives of the study were clarified. The evaluators went over the task list and clarified any doubts they had. The evaluators performed the task assigned using the website assigned and recorded all answers to the questions in the sheets given.

Reporting – the results included the responses to the questions. Analysis of the results was conducted in the next section.

Guideline Development

The data extracted in the previous step were analyzed to create usability guidelines. The results of the questionnaire included a large set of qualitative data. A card sorting exercise was conducted to categorize the responses.

Card Sorting

Card sorting is a knowledge elicitation technique that involves categorizing items into groups using a single criterion (Bussolon, Russi, & Missier, 2006). Items are grouped by participant’s choice, and they provide a name for the group (Nawaz, 2012). This procedure is repeated until it is no longer feasible to do so. This technique is known for its simplicity, focus-oriented, and ability to elicit knowledge (Zimmermann, 2016). There are two methods of card sorting open and closed. In an open card sorting, the participants are given cards with no pre-established groupings and are asked to sort that and name the groups. In closed card sorting, participants are given cards with initial groups and are required to place the card into pre-established groups. In this study, an open card sorting exercise was conducted.

The card sorting activity was organized as follows;

Participants – six participants who have expertise in web development were selected. An open invitation was sent to suitable candidates, and they were asked to volunteer for this exercise. It was mandatory for the participants to have an understanding of usability guidelines.

Card – the cards were prepared that have the responses written from the questionnaire. A single response was listed on each card.

Procedure – open card sorting exercise was conducted in a two-hour session organized with the participants. The process was repeated several times before deriving the final set of categories.

The card sorting exercise resulted in the following ten groups; website organization, website navigation, website controls, images, feedback, task, item display, and item search. Figure 3 illustrates the mapping of responses to groups. The responses were converted into guidelines. Table 2 lists the mobile-first usability guideline.

Figure 3. Mapping diagram

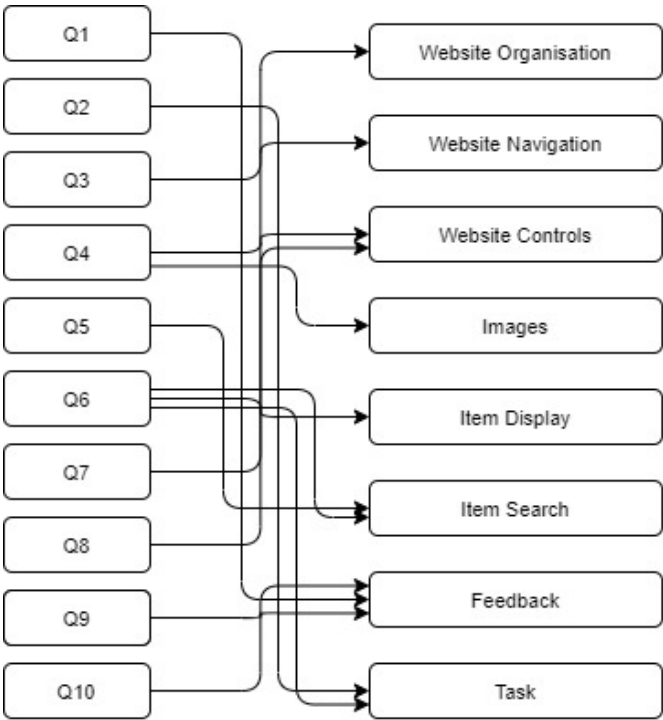


Table 2. Mobile-first usability guideline

Web Site Organization
G1.Provide Index page
G2.Minimize the number of clicks
G3.Open new page in new window
G4.Eliminate horizontal scrolling
Web Site Navigation
G5.Make navigation evident from the first page
G6.Navigation should be easy to find
G7.Provide navigate option to move between pages
G8.Use standard navigation Patten
Web Site Controls
G9. Use short names for buttons
G10.Minimize the number of buttons on a page
G11.Controls must be large enough for touch
G12.Visullay align controls.
G13. Use consistent and meaningful icons
Images
G14.Use optimized images
G15.Provide zoom in and zoom out features
Feedback
G16.Provide feedback while users are waiting
G17.Use help to assist customers
Task
G18.All tasks must be sequentially organized
G19.Standardize common task sequence
Item Display
G20.Ensure only necessary information is displayed
G21.Product information should be in homogeneous chunk
G22.Visited products denoted by color change
Item search
G23.Provide search field
G24.Search field should be on the top
G25.Provides search history
G26.Provide auto-complete option

Guideline Validation

The guideline was validated to ensure that it will provide a usable experience. In literature, there is no standard framework to validate usability guidelines. In previous work, Kumar et al. (2019) validated the usability guidelines through usability assessment. A similar approach has been adopted in this

paper. Heuristic evaluation, a common usability assessment method for evaluating user interfaces, was used to validate the guideline. Heuristic evaluation can with ease, identify a large number of usability defects in a user interface (Jimenez, Lozada, & Rosas, 2016; Kumar & Goundar, 2019). This method also requires little preparation and is also cost-effective (Kumar, Goundar, & Chand, 2020; Tondello, Kappen, Mekler, Ganaba, & Nacke, 2016).

Experiment Setup

The experiment setup was as follows;

Participants – Nielsen and Molich (1990) recommended that 3-5 evaluators should be selected who have expertise in the area of human-computer interaction. Based on the recommendation, three usability experts were selected.

Hardware and Software – prototypes of the responsive e-commerce user interfaces were designed using the proposed guideline. Hardware devices such as smartphones, tablet pc, and Ipad were used for the evaluation.

Documents and Material – necessary documents to support the evaluation process were prepared. A list of heuristics initially proposed by Nielsen and Molich (1990) was used. Table 3 provides a list of heuristics used for evaluation.

Table 3. List of Heuristics

ID	Heuristic
H ₁	Visibility of System Status
H ₂	Match between system and the real world
H ₃	User control and freedom
H ₄	Consistency and standards
H ₅	Error prevention
H ₆	Recognition rather than recall
H ₇	Flexibility and efficiency of use
H ₈	Aesthetic and minimalist design
H ₉	Help users recognize, diagnose, and recover from errors
H ₁₀	Help and Documentation

The Likert ranking scale proposed by (Dimalopoulos & Magoulas, 2009) was used;

- 1- Usability Disaster,
- 2- Major Problem,
- 3- Minor Problem,
- 4- Superficial Problem
- 5- Not A Problem

Procedure – the heuristic evaluation was conducted by the experts. The experts were explained the goal of the evaluation and were asked to clarify any doubt they had. The heuristic evaluation was conducted, and the experts recorded the severity ratings for each of the heuristics. Finally, a debriefing session was conducted to reflect on their experience while conducting the heuristic evaluation.

RESULTS

The primary data from the heuristic evaluation were severity ratings. The evaluator ratings were recorded in a worksheet and finally collated to make a summary. The results mainly highlighted the cases detailing the location of the problem and its associated heuristic. The prototypes were generally appreciated by the evaluators. For the majority of the heuristics, average ratings included 'superficial' or 'not a problem'. The evaluators felt that most of the heuristics had been fully implemented. However, they indicated minor usability problems in H9 and H10. For H9, it was difficult to demonstrate using prototypes how users will diagnose and recover from errors. Another area of concern was the help feature which was categorized under H10, and the evaluators believe that a help feature should be available. The usability concerns that were raised H9, H10 were mainly the limitations of using prototypes.

Limitations

Few limitations may affect the generalization of our results. Firstly, only five websites were used to extract information about good usability practices. More websites can be evaluated in the future, and the proposed guideline can be updated. Secondly, responsive e-commerce websites need to be fully developed using the proposed guideline and rigorously evaluated to provide a better generalization of the results.

CONCLUSION AND FUTURE WORK

Recently there has been increased interest in the use of responsive web design approach to build electronic commerce websites that allow them to be easily viewed on different devices with different screen dimensions (Maurois & Amanda, 2018). Usability is a major challenge for the developers to design applications that can provide an efficient user experience across different devices (Bernacki et al., 2016; Majid, Kamaruddin, & Mansor, 2015). In this paper, a mobile-first usability guideline was developed to assist in designing responsive e-commerce web applications. There is a lack of an accepted method for designing usability guidelines. In previous studies, Kumar et al. (2019) designed a usability guideline for mobile learning applications by extracting usability problems from mobile learning studies reported in the literature. Gumussoy (2016) designed a usability guideline for internet banking by analyzing the available usability problems in the domain. The lack of available usability studies on responsive e-commerce websites led us to follow a different approach. In this study, good practices from five responsive e-commerce websites were extracted to formulate the usability guideline.

A systematic approach was adopted to design and validate the mobile-first usability guideline. Five popular responsive e-commerce websites were selected based on the assumption that they provide a pleasant user experience. Nielsen and Molich (1990) designed ten heuristics that is widely accepted in evaluating user interfaces, and websites that conform to these usability heuristics provide a high level of user satisfaction. These heuristics were translated into questionnaires to guide the information extraction from the websites. Ten usability experts performed the task assigned using the websites and answered the questions. The results of the questionnaire provided us with a large set of qualitative data that was analyzed using a card sorting exercise. The card sorting exercise resulted in eight groups. The responses within each of the groups were converted to usability guidelines.

Finally, the usability guideline was validated by conducting a heuristic evaluation, and the results indicate it will have a positive impact on user satisfaction. The usability guidelines will be beneficial in the following ways; i) it will provide a usable experience, ii) help to achieve standardization in the design of responsive e-commerce websites, iii) fewer user errors will save cost from rework and, iv) assist developers in achieving shorter time to market the product. This study also sets the direction for further investigation in the field. Future research may be conducted to extract data from more websites and updated the usability guideline. More responsive e-commerce websites need to be developed using the proposed guideline and rigorously evaluated to better generalize the results.

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CONFLICT OF INTEREST

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Bimal Aklesh Kumar is a lecturer at Fiji National University in the department of Computer Science and Information Systems. He graduated from University of the South Pacific with BSc, MSc and is currently a PhD candidate at the University of South Pacific. His research area includes software engineering, mobile computing, and e-commerce. He has published several research articles in this field.